Lecture 11: Sections 3.2 and 3.3

Derivatives of Logarithmic, Exponential, and Inverse Functions

Md Ahsanul Haque, Ph.D.

Assistant Professor
Department of Mathematics and Physics
North South University, Dhaka

Derivatives of Logarithmic Functions

Theorem (Basic Logarithmic Derivative)

$$\frac{d}{dx}[\ln x] = \frac{1}{x}, \quad x > 0$$

Theorem (General Logarithmic Derivative)

If u is a differentiable function of x and $u > 0$, then:

$$\frac{d}{dx}[\ln u] = \frac{1}{u} \cdot \frac{du}{dx}$$

Example

1. $\frac{d}{dx}[\ln(3x^2+1)] = \frac{1}{3x^2+1} \cdot 6x = \frac{6x}{3x^2+1}$
2. $\frac{d}{dx}[\ln(\sin x)] = \frac{1}{\sin x} \cdot \cos x = \cot x$

Logarithmic Differentiation Technique

When to Use Logarithmic Differentiation

- Products of multiple functions: $f(x)g(x)h(x)$
- Quotients with complex numerators/denominators
- Functions with variables in both base and exponent
- Complicated powers: $[f(x)]^{g(x)}$

Method

1. Take the natural log of both sides: $\ln y = \ln f(x)$
2. Use log properties to simplify
3. Differentiate implicitly
4. Solve for $\frac{dy}{dx}$
5. Substitute back for y

Problem 1: Logarithmic Differentiation

Find the derivative of $y = x\sqrt{\frac{x^2+1}{x^2-1}}$

Solution 1: Logarithmic Differentiation

1. Take natural log: $\ln y = \ln\left(x\sqrt{\frac{x^2+1}{x^2-1}}\right)$
2. Simplify using log properties:
$$\ln y = \ln x + \frac{1}{2}\ln(x^2+1) - \frac{1}{2}\ln(x^2-1)$$
3. Differentiate implicitly:
$$\frac{1}{y}\cdot\frac{dy}{dx} = \frac{1}{x} + \frac{1}{2}\cdot\frac{2x}{x^2+1} - \frac{1}{2}\cdot\frac{2x}{x^2-1}$$
$$\frac{1}{y}\cdot\frac{dy}{dx} = \frac{1}{x} + \frac{x}{x^2+1} - \frac{x}{x^2-1}$$
4. Solve for $\frac{dy}{dx}$:
$$\frac{dy}{dx} = y\left(\frac{1}{x} + \frac{x}{x^2+1} - \frac{x}{x^2-1}\right)$$
5. Substitute back:
$$\frac{dy}{dx} = x\sqrt{\frac{x^2+1}{x^2-1}}\left(\frac{1}{x} + \frac{x}{x^2+1} - \frac{x}{x^2-1}\right)$$

Problem 2: Variable Exponents

Find the derivative of $y = x^x$

Solution 2: Variable Exponents

1. Take natural log:
$$\ln y = \ln(x^x) = x \ln x$$
2. Differentiate implicitly:
$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln x + x \cdot \frac{1}{x} = \ln x + 1$$
3. Solve for $\frac{dy}{dx}$:
$$\frac{dy}{dx} = y(\ln x + 1) = x^x(\ln x + 1)$$

General Formula

For $y = [f(x)]^{g(x)}$:

$$\frac{dy}{dx} = [f(x)]^{g(x)} \left[g'(x) \ln f(x) + g(x) \frac{f'(x)}{f(x)} \right]$$

Problem 3: Complex Product

Find the derivative of $y = \dfrac{(x+1)^2(x-2)^3}{(x+3)^4}$

Solution 3: Complex Product

1. Take natural log:

$$\ln y = 2\ln(x+1) + 3\ln(x-2) - 4\ln(x+3)$$

2. Differentiate implicitly:

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{2}{x+1} + \frac{3}{x-2} - \frac{4}{x+3}$$

3. Solve for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = y\left(\frac{2}{x+1} + \frac{3}{x-2} - \frac{4}{x+3}\right)$$

$$\frac{dy}{dx} = \frac{(x+1)^2(x-2)^3}{(x+3)^4}\left(\frac{2}{x+1} + \frac{3}{x-2} - \frac{4}{x+3}\right)$$

Practice: Logarithmic Differentiation

Find the derivatives using logarithmic differentiation:

1. $y = \sqrt[3]{\frac{x(x^2+1)}{\sqrt{x^2-1}}}$
2. $y = (\sin x)^x$
3. $y = \frac{x^2\sqrt[3]{2x-1}}{(x+1)^2}$
4. $y = x^{\ln x}$
5. $y = (1+\frac{1}{x})^x$

Derivative of Inverse Functions

Theorem (Derivative of Inverse Function)

If f is one-to-one and differentiable with inverse f^{-1}, then:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

provided $f'(f^{-1}(x)) \neq 0$.

A function is one-to-one if it is either:

- Strictly increasing on its domain, or
- Strictly decreasing on its domain

Example

If $f(x) = x^3 + 2x + 1$ and $f(1) = 4$, then $f^{-1}(4) = 1$ and:

$$(f^{-1})'(4) = \frac{1}{f'(1)} = \frac{1}{3(1)^2 + 2} = \frac{1}{5}$$

Derivatives of Exponential Functions

Theorem (Basic Exponential Derivative)

$$\frac{d}{dx}[e^x] = e^x$$

Theorem (General Exponential Derivative)

If u is a differentiable function of x, then:

$$\frac{d}{dx}[e^u] = e^u \cdot \frac{du}{dx}$$

Theorem (General Exponential Base)

$$\frac{d}{dx}[a^x] = a^x \ln a, \quad a > 0$$

$$\frac{d}{dx}[a^u] = a^u \ln a \cdot \frac{du}{dx}$$

Problem 4: Exponential Derivatives

Find the derivatives:

1. $y = e^{x^2-3x}$
2. $y = 2^{3x}$
3. $y = x^2 e^{-x}$

Solution 4: Exponential Derivatives

1. $y = e^{x^2-3x}$
   $\frac{dy}{dx} = e^{x^2-3x} \cdot (2x-3)$
2. $y = 2^{3x}$
   $\frac{dy}{dx} = 2^{3x} \ln 2 \cdot 3 = 3 \cdot 2^{3x} \ln 2$
3. $y = x^2 e^{-x}$ (Product Rule)
   $\frac{dy}{dx} = 2xe^{-x} + x^2(-e^{-x}) = e^{-x}(2x - x^2)$

Derivatives of Inverse Trigonometric Functions

Theorem (Inverse Trigonometric Derivatives)

$$\frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}[\cos^{-1} x] = -\frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}[\tan^{-1} x] = \frac{1}{1+x^2}$$

$$\frac{d}{dx}[\cot^{-1} x] = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}[\sec^{-1} x] = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

$$\frac{d}{dx}[\csc^{-1} x] = -\frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

Problem 5: Inverse Trigonometric Derivatives

Find the derivatives:

1. $y = \sin^{-1}(2x)$
2. $y = \tan^{-1}(\sqrt{x})$
3. $y = x \cos^{-1} x$

Solution 5: Inverse Trigonometric Derivatives

1. $y = \sin^{-1}(2x)$
   $\frac{dy}{dx} = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2 = \frac{2}{\sqrt{1-4x^2}}$
2. $y = \tan^{-1}(\sqrt{x})$
   $\frac{dy}{dx} = \frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}(1+x)}$
3. $y = x\cos^{-1} x$ (Product Rule)
   $\frac{dy}{dx} = \cos^{-1} x + x \cdot \left(-\frac{1}{\sqrt{1-x^2}}\right) = \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$

Problem 6: Combined Functions

Find the derivative of $y = \ln(\sin^{-1} x)$

Solution 6: Combined Functions

1. Chain rule with logarithmic and inverse trig:

$$\frac{dy}{dx} = \frac{1}{\sin^{-1} x} \cdot \frac{d}{dx}[\sin^{-1} x]$$

2. Apply inverse trig derivative:

$$\frac{dy}{dx} = \frac{1}{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}}$$

3. Final answer:

$$\frac{dy}{dx} = \frac{1}{\sin^{-1} x \cdot \sqrt{1-x^2}}$$

Problem 7: Implicit with Exponential

Find $\frac{dy}{dx}$ for $e^{xy} = x + y$

Solution 7: Implicit with Exponential

1. Differentiate both sides: $\frac{d}{dx}[e^{xy}] = \frac{d}{dx}[x + y]$
2. Apply chain rule and product rule:

$$e^{xy} \cdot (y + x\frac{dy}{dx}) = 1 + \frac{dy}{dx}$$

3. Expand:

$$ye^{xy} + xe^{xy}\frac{dy}{dx} = 1 + \frac{dy}{dx}$$

4. Collect $\frac{dy}{dx}$ terms:

$$xe^{xy}\frac{dy}{dx} - \frac{dy}{dx} = 1 - ye^{xy}$$

5. Solve:

$$\frac{dy}{dx}(xe^{xy} - 1) = 1 - ye^{xy}$$

$$\frac{dy}{dx} = \frac{1 - ye^{xy}}{xe^{xy} - 1}$$

Advanced Practice Problems

1. Find $\frac{dy}{dx}$ for $y = \ln\left(\frac{\sqrt{x^2+1}}{x\sqrt[3]{2x-1}}\right)$
2. Differentiate $y = (\cos x)^{\sin x}$
3. Find the derivative of $y = \sec^{-1}(e^{2x})$
4. Find $\frac{dy}{dx}$ for $\ln(x^2 + y^2) = 2\tan^{-1}(\frac{y}{x})$
5. Differentiate $y = x^{\sin^{-1} x}$ using logarithmic differentiation
6. Find the derivative of $y = \frac{e^x \sin^{-1} x}{\sqrt{1+x^2}}$

Summary of Key Formulas

Logarithmic and Exponential Derivatives

$$\frac{d}{dx}[\ln x] = \frac{1}{x}$$

$$\frac{d}{dx}[\ln u] = \frac{1}{u} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}$$

$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[e^u] = e^u \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[a^x] = a^x \ln a$$

$$\frac{d}{dx}[a^u] = a^u \ln a \cdot \frac{du}{dx}$$

Summary of Key Formulas (Continued)

Inverse Trigonometric Derivatives

$$\frac{d}{dx}[\sin^{-1} u] = \frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[\cos^{-1} u] = -\frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[\tan^{-1} u] = \frac{1}{1+u^2} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[\cot^{-1} u] = -\frac{1}{1+u^2} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[\sec^{-1} u] = \frac{1}{|u|\sqrt{u^2-1}} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[\csc^{-1} u] = -\frac{1}{|u|\sqrt{u^2-1}} \cdot \frac{du}{dx}$$

When to Use Logarithmic Differentiation

Use When You See:

- Products: $f(x)g(x)h(x)$
- Quotients: $\frac{\text{complex numerator}}{\text{complex denominator}}$
- Variable exponents: $[f(x)]^{g(x)}$
- Multiple layers of composition

Advantages

- Turns products into sums
- Turns quotients into differences
- Turns powers into products
- Simplifies complex functions

Remember

Always check if logarithmic differentiation will actually simplify your problem before using it!